Vikash Vashisth

CFD Engineer | Thermal-Fluid and Multiphysics Simulation

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Work authorization: Authorized to work in Sweden. No employer sponsorship required to start employment.

PROFESSIONAL SUMMARY

CFD engineer with a PhD in Chemical Engineering and experience in academic and industry-linked R&D. Uses ANSYS Fluent, STAR-CCM+ and Python automation to evaluate thermal-fluid performance and compare equipment designs. Combines numerical verification, experimental validation and physical interpretation to support engineering decisions.

CORE EXPERTISE

Applications: Fluid dynamics; thermal management and conjugate heat transfer; reacting and multiphase flows; rotating machinery; non-Newtonian flows.

Methods: RANS/LES; VOF; finite-rate chemistry; sliding mesh; adaptive refinement; mesh independence; convergence and conservation checks; experimental validation.

Tools: ANSYS Fluent; STAR-CCM+; HFSS; Python, PyFluent and PyAEDT; MATLAB; UDF/C; Siemens NX; ANSYS meshing and DesignModeler; HPC.

PROFESSIONAL EXPERIENCE

Chalmers University of Technology — Gothenburg, Sweden

Researcher - CFD and Multiphysics — Dec 2025 - Present

Postdoctoral Researcher - CFD and Multiphysics — Dec 2022 - Dec 2025

- Evaluated microwave plasma-reactor performance by developing 3D CFD models of coupled flow, heat transfer and nitrogen-fixation chemistry; validated predictions against experimental measurements.
- Evaluated forced-jet quenching to limit downstream NO destruction. Compared with the intrinsic-quenching baseline, CFD predicted 6-10% higher NO yield and 6-8% lower specific energy consumption.
- Developed fully automated two-way HFSS-Fluent coupling using PyAEDT and PyFluent, integrating electromagnetic power deposition with fluid flow, heat transfer and finite-rate chemistry.
- Investigated turbulent bubble and droplet breakup and fragment formation; designed and ran LES-VOF simulations with adaptive mesh refinement to resolve interface dynamics and fragment-size distributions.
- Verified LES resolution and numerical sensitivity, and validated breakup predictions against high-speed experimental measurements.
- Led biweekly CFD reviews with NitroCapt AB and research partners, explaining model results and presenting engineering recommendations.

IIT Guwahati - Reckitt Benckiser Collaboration — Guwahati, India

Research Associate - Industrial CFD Project — Jan 2022 - Apr 2022

- Investigated shear-thinning liquid spreading for the Reckitt Benckiser collaboration; built transient 3D VOF models using experimentally measured rheology.
- Established mesh-independent predictions and validated liquid-spreading behaviour against experimental measurements.

IIT Roorkee — Roorkee, India

Doctoral Researcher - Fluid Mechanics, Rotating Machinery and Mixing — Dec 2015 - Apr 2021

- Evaluated how rotor-stator geometry affects mixing and power demand using sliding-mesh RANS and LES; assessed flow distribution, mixing performance and mesh independence.

- Benchmarked predicted power demand for conventional and newly developed mixer configurations; power numbers agreed within approximately 10% of measurements.

SELECTED ENGINEERING PROJECT

Electric Drive Unit Cooling Jacket - Conjugate Heat Transfer 2026

Independent engineering simulation and geometry-comparison study

- Evaluated liquid cooling of an electric drive unit; built a STAR-CCM+ conjugate heat-transfer model coupling coolant flow with conduction through the housing and equivalent stator.
- Established mesh-independent CFD predictions and evaluated cooling-channel geometry variations.
- Evaluated cooling gains against pressure losses from additional channel turns. At 0.110 kg/s, seven versus six turns reduced predicted peak equivalent-stator temperature by 1.4 K and increased pressure loss by 16.2%.

EDUCATION

Indian Institute of Technology Roorkee Roorkee, India

PhD, Chemical Engineering 2015 - 2021

M.Tech, Chemical Engineering 2011 - 2013

SELECTED PUBLICATIONS

2026 | Forced and intrinsic quenching for nitrogen fixation. Chemical Engineering Journal.

2025 | Dynamics of bubble breakup under turbulent flow conditions. Chemical Engineering Journal.

2024 | Plasma reactor performance for direct nitrogen fixation. Chemical Engineering Journal.

TRAINING & ADDITIONAL INFORMATION

Training: Ansys Learning Hub - PyFluent and HFSS (2026).

Languages and mobility: English - fluent; Swedish - A2 certificate, continuing studies; Swedish driving licence (B); open to relocation.